

NammaPG – User Entity & ORM Basics

1. What is ORM?

ORM (Object Relational Mapping) is a technique that allows developers to map Java objects directly to database tables. In Spring Boot, Hibernate is the most commonly used ORM framework.

2. Why ORM is used in NammaPG

- No manual SQL table creation
- Schema managed via Java code
- Database-agnostic design
- Cleaner, scalable architecture

3. Technology Stack Used

- Java 17 (LTS)
- Spring Boot 3.5.x
- Hibernate ORM 6.x
- MySQL 8.0
- application.yml for configuration

4. User Entity Purpose

The User entity represents all types of users in the NammaPG system, including tenants, PG owners, administrators, and volunteers.

5. User Entity Fields

- id – Primary key
- fullName – User's full name
- email – Unique identifier for login
- password – Encrypted password
- active – Account status
- createdAt – Record creation timestamp
- updatedAt – Last update timestamp

6. Important JPA Annotations Explained

@Entity – Marks class as a database table

@Table – Defines table name

@Id – Primary key

@GeneratedValue – Auto-increment ID

@Column – Column constraints

@PrePersist – Before insert logic

@PreUpdate – Before update logic

7. Hibernate Schema Auto-Creation

With 'spring.jpa.hibernate.ddl-auto=update', Hibernate automatically:

- Creates tables
- Adds columns
- Updates schema safely (dev mode)

8. Verification

The users table was successfully created in MySQL using Hibernate, confirming correct ORM setup.

9. Next Steps

- Role entity creation

- User-Role mapping (Many-to-Many)
- Authentication & authorization schema